

Advanced (Habanero)

Q1. What is the primary purpose of a box-and-whisker plot?

- (A) To display the mean and median of a dataset
- (B) To show the distribution of data and detect outliers
- (C) To calculate the standard deviation of a dataset
- (D) To create a histogram
- (E) To find the mode of a dataset

Q2. Which part of a box-and-whisker plot represents the interquartile range (IQR)?

- (A) The whiskers
- (B) The box
- (C) The median line
- (D) The outliers
- (E) The scale

Q3. What is the formula to calculate the 3rd quartile (Q_3)?

- (A) $Q_3 = \frac{3(n+1)}{4}$

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Q4. Which of the following is NOT a characteristic of a box-and-whisker plot?

- (A) It displays the median and quartiles
- (B) It shows the distribution of data
- (C) It is used to compare multiple datasets
- (D) It is a type of histogram
- (E) It is used to detect outliers

Q5. What does the 1st quartile (Q_1) represent in a box-and-

whisker plot?

- (A) The median of the dataset
- (B) The lower 25
- (B) The upper 25
- (B) The interquartile range (IQR)
- (C) The range of the dataset

Q6. Which of the following datasets would have the largest interquartile range (IQR)?

- (A) 1, 2, 3, 4, 5
- (B) 1, 1, 1, 1, 1
- (C) 10, 20, 30, 40, 50
- (D) 100, 200, 300, 400, 500
- (E) 1, 3, 5, 7, 9

Q7. How do you calculate the 1st quartile (Q_1) in a dataset of n values?

- (A) $Q_1 = \frac{n+1}{4}$

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Q8. A value is considered an outlier if it is $1.5 \cdot IQR$ away from the nearest quartile. What does IQR represent?

- (A) Interquartile range
- (B) Inverse quartile range
- (C) Interval quartile range
- (D) Internal quartile range
- (E) Intriquartile range

Open Ended Questions (FRQ)

Q9. Construct a box-and-whisker plot for the following dataset: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20. Identify the median, 1st and 3rd quartiles, and any outliers. Show your work and explain your reasoning.

Q10. Compare and contrast a box-and-whisker plot with a histogram. How do they differ in terms of the information they convey and their uses in data analysis? Provide examples to support your answer.

Chef's Tips

- Provide clear instructions and examples for each question
- Allow students to use calculators and statistical software
- Emphasize the importance of showing work and explaining reasoning